Ecuador rCARI Validation Study Primary Results

Nicole Wu

November 19, 2023

1 Survey Progress

Table 1: Sample Completion: Face-to-Face vs. Remote

	Not Completed		Completed		Total	
	Count	Percent	Count	Percent	Count	Percent
Face-To-Face	413	35.21	760	64.79	1173	100.00
Remote	543	45.29	656	54.71	1199	100.00
Total	956	40.30	1416	59.70	2372	100.00

Figure 1: Data Collection Progress by Date and Modality

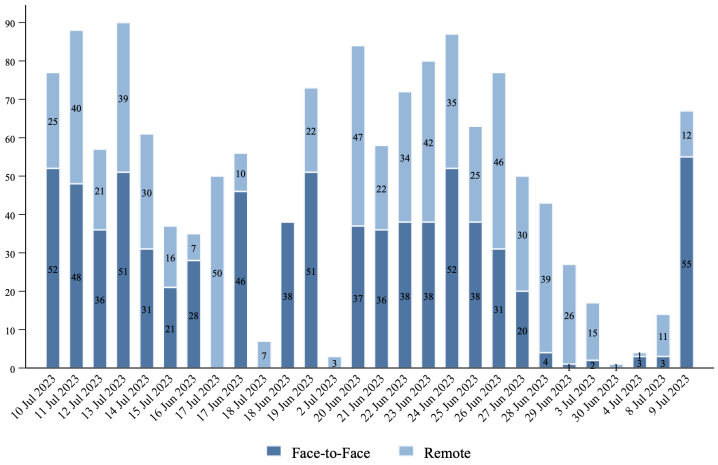


Figure 2: F2F Data Collection Progress by Date and Enumerator

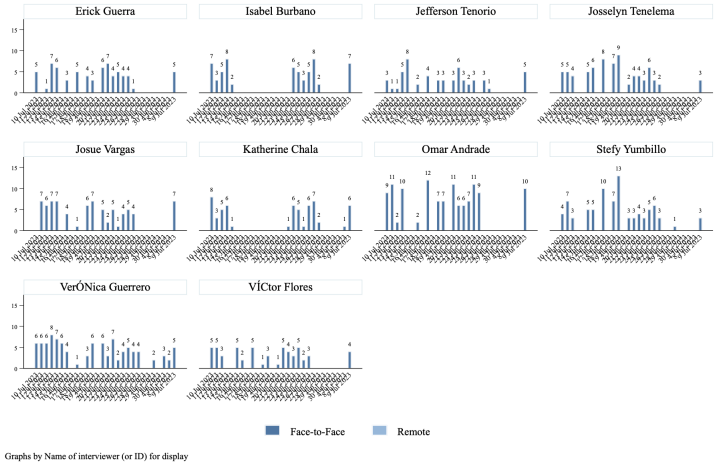


Figure 3: Remote Data Collection Progress by Date and Enumerator

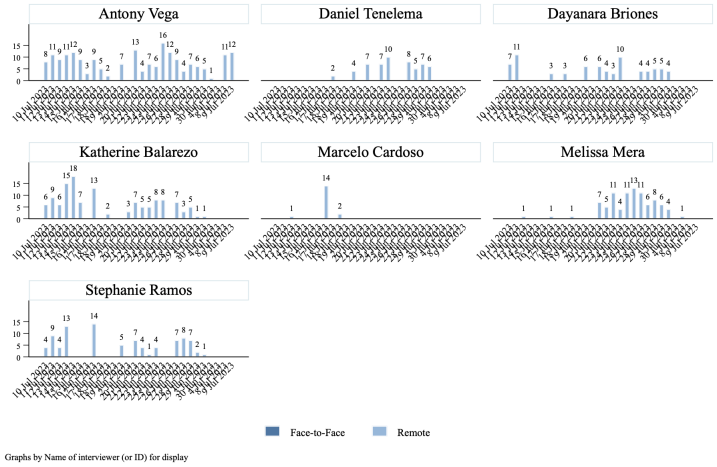
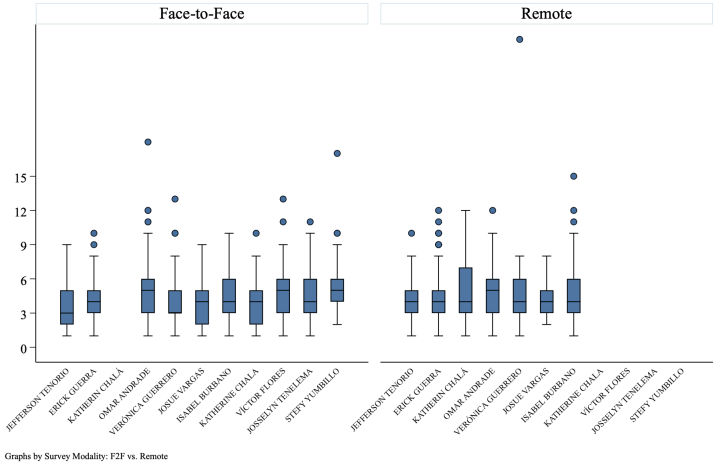


Figure 4: Household Size by Modality and Enumerator



2 Randomization Quality

To assess the balance of the treatment (Remote) and control (F2F) groups, we examined various respondent and household characteristics, including basic demographics, income sources, and changes over time. This allowed us to evaluate whether the two groups were similar in terms of these factors, ensuring a fair comparison between the treatment and control conditions.

2.1 Modality Randomization

Table 2: Balance Table on Randomized Survey Modality Assignment: Face-to-Face vs. Remote

		(1) F2F		(2) Remote		(1)-(2) Pairwise t-test
Variable	N	Mean/(SE)	N	Mean/(SE)	N	Mean difference
Respondent Age	556	46.19	656	43.00	1212	3.19***
		(0.76)		(0.66)		
Female Respondent	760	0.77	656	0.76	1416	0.01
		(0.02)		(0.02)		
Head of Household Respondent	760	0.61	656	0.53	1416	0.08***
		(0.02)		(0.02)		
Household Size	760	4.37	656	4.57	1416	-0.19
		(0.08)		(0.09)		
How many household members are contributing to the household income?	759	1.47	655	1.45	1414	0.01
		(0.03)		(0.03)		
How many income sources does your household have?	760	1.40	656	1.50	1416	-0.11***
		(0.03)		(0.03)		
Income Source: None	760	0.04	656	0.02	1416	0.02**
		(0.01)		(0.00)		
Income Source: Skilled Waged Labor	760	0.08	656	0.06	1416	0.02
		(0.01)		(0.01)		
Income Source: Unskilled Waged Labor	760	0.35	656	0.37	1416	-0.02
		(0.02)		(0.02)		
Income Source: Trade	760	0.15	656	0.18	1416	-0.02
		(0.01)		(0.01)		
Income Source: Agricultural Production	760	0.04	656	0.10	1416	-0.06***
		(0.01)		(0.01)		
Income Change: None	760	0.49	656	0.52	1416	-0.03
		(0.02)		(0.02)		
Income Change: Increase	760	0.03	656	0.04	1416	-0.01
		(0.01)		(0.01)		
Income Change: Reduced < 25%	760	0.17	656	0.13	1416	0.04**
		(0.01)		(0.01)		
Income Change: Reduced 25% - 50%	760	0.14	656	0.13	1416	0.02
		(0.01)		(0.01)		
Income Change: Reduced 50+%	760	0.16	656	0.18	1416	-0.02
		(0.01)		(0.02)		

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

2.2 F2F Length Randomization

Table 3: Balance Table on Randomized Survey Length Assignment in F2F Survey: Long vs. Short

Variable	N	(1) F2F Mean/(SE)	N	(2) Remote Mean/(SE)	N	(1)-(2) Pairwise t-test Mean difference
Respondent Age	268	47.25	288	45.22	556	2.03
		(1.08)		(1.07)		
Female Respondent	377	0.75	383	0.79	760	-0.04
		(0.02)		(0.02)		
Head of Household Respondent	377	0.62	383	0.61	760	0.01
		(0.03)		(0.03)		
Single Respondent	377	0.24	383	0.26	760	-0.02
		(0.02)		(0.02)		
Married Respondent	377	0.46	383	0.45	760	0.00
		(0.03)		(0.03)		
IDP Household	377	0.00	383	0.01	760	-0.00
		(0.00)		(0.00)		
Household has disabled member	377	0.32	383	0.21	760	0.11***
		(0.02)		(0.02)		
Number of disabled household members	377	0.36	383	0.23	760	0.13***
		(0.03)		(0.02)		
Household head has attended school	377	0.85	383	0.87	760	-0.03
		(0.02)		(0.02)		
Household Size	377	4.30	383	4.44	760	-0.14
		(0.12)		(0.11)		
Total number of male household member	377	2.12	383	2.16	760	-0.03
		(0.08)		(0.08)		
Total number of female household member	377	2.20	383	2.31	760	-0.10
		(0.07)		(0.07)		
Total number of children	377	1.62	383	1.75	760	-0.13
		(0.08)		(0.08)		
Total number of senior	377	0.51	383	0.52	760	-0.01
		(0.04)		(0.04)		
Total number of adults	377	2.19	383	2.19	760	-0.00
		(0.07)		(0.08)		
Household Female/Male Ratio	377	1.68	383	1.75	760	-0.06
		(0.09)		(0.10)		
Household Dependency Ratio	377	1.65	383	1.72	760	-0.07
		(0.09)		(0.09)		
How many household members are contributing to the household income?	376	1.48	383	1.46	759	0.02
		(0.04)		(0.04)		
How many income sources does your household have?	377	1.40	383	1.40	760	0.00
		(0.04)		(0.04)		
Income Source: None	377	0.05	383	0.03	760	0.02
		(0.01)		(0.01)		
Income Source: Skilled Waged Labor	377	0.06	383	0.09	760	-0.03
		(0.01)		(0.01)		
Income Source: Unskilled Waged Labor	377	0.32	383	0.38	760	-0.05
		(0.02)		(0.02)		
Income Source: Trade	377	0.18	383	0.13	760	0.04*
		(0.02)		(0.02)		
Income Source: Agricultural Production	377	0.05	383	0.04	760	0.01
		(0.01)		(0.01)		
Income Change: None	377	0.48	383	0.50	760	-0.02
		(0.03)		(0.03)		
Income Change: Increase	377	0.03	383	0.03	760	-0.00
		(0.01)		(0.01)		
Income Change: Reduced < 25%	377	0.18	383	0.17	760	0.01
		(0.02)		(0.02)		
Income Change: Reduced 25% - 50%	377	0.15	383	0.14	760	0.01
		(0.02)		(0.02)		
Income Change: Reduced 50+%	377	0.16	383	0.16	760	-0.00
		(0.02)		(0.02)		

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

2.3 Remote Length Randomization

Table 4: Balance Table on Randomized Survey Length Assignment in Remote Survey: Long vs. Short

Variable	N	(1) F2F Mean/(SE)	N	(2) Remote Mean/(SE)	N	(1)-(2) Pairwise t-test Mean difference
Respondent Age	324	43.37	332	42.64	656	0.73
		(0.95)		(0.91)		
Female Respondent	324	0.74	332	0.77	656	-0.03
		(0.02)		(0.02)		
Head of Household Respondent	324	0.52	332	0.54	656	-0.02
		(0.03)		(0.03)		
Household Size	324	4.64	332	4.50	656	0.14
		(0.14)		(0.12)		
How many household members are contributing to the household income?	323	1.46	332	1.45	655	0.01
		(0.04)		(0.04)		
How many income sources does your household have?	324	1.52	332	1.48	656	0.04
		(0.05)		(0.04)		
Income Source: None	324	0.02	332	0.01	656	0.01
		(0.01)		(0.01)		
Income Source: Skilled Waged Labor	324	0.06	332	0.06	656	-0.00
		(0.01)		(0.01)		
Income Source: Unskilled Waged Labor	324	0.38	332	0.37	656	0.01
		(0.03)		(0.03)		
Income Source: Trade	324	0.19	332	0.17	656	0.02
		(0.02)		(0.02)		
Income Source: Agricultural Production	324	0.11	332	0.09	656	0.02
		(0.02)		(0.02)		
Income Change: None	324	0.54	332	0.50	656	0.03
		(0.03)		(0.03)		
Income Change: Increase	324	0.04	332	0.03	656	0.01
		(0.01)		(0.01)		
Income Change: Reduced < 25%	324	0.11	332	0.15	656	-0.04
		(0.02)		(0.02)		
Income Change: Reduced 25% - 50%	324	0.12	332	0.13	656	-0.01
		(0.02)		(0.02)		
Income Change: Reduced 50+%	324	0.19	332	0.18	656	0.01
		(0.02)		(0.02)		

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

3 Primary Results

3.1 Impact of Survey Modality on Key Food Security Outcomes

Table 5: Impact of Survey Modality: Face-to-Face vs. Remote

Variable	(1) F2F Mean/(SE)	(2) Remote Mean/(SE)	(1)-(2) Pairwise t-test Mean difference
Consumption over the past 7 days (cereals and tubers)	5.38	4.93	0.45***
	(0.08)	(0.09)	
Consumption over the past 7 days (pulses)	2.00	2.50	-0.50***
	(0.07)	(0.09)	
Consumption over the past 7 days (dairy products)	1.38	2.03	-0.65***
	(0.07)	(0.09)	
Consumption over the past 7 days (protein-rich foods)	3.03	3.48	-0.44***
	(0.09)	(0.09)	
Consumption over the past 7 days (vegetables)	3.93	4.75	-0.82***
	(0.10)	(0.10)	
Consumption over the past 7 days (fruit)	2.58	2.90	-0.31**
	(0.09)	(0.10)	
Consumption over the past 7 days (oil)	5.38	5.40	-0.02
	(0.09)	(0.09)	
Consumption over the past 7 days (sugar)	5.39	5.64	-0.25*
	(0.09)	(0.09)	
Food Consumption Score	46.28	52.54	-6.25***
	(0.78)	(0.94)	
Relied on less preferred, less expensive food	4.41	4.56	-0.15
	(0.09)	(0.11)	
Borrowed food or relied on help from friends or relatives	1.64	1.75	-0.11
	(0.07)	(0.08)	
Reduced portion size of meals at meals time	3.80	3.27	0.53***
	(0.10)	(0.11)	
Restricted Coping by adults in order for young-children to eat	1.87	1.51	0.36***
	(0.09)	(0.09)	
Reduced the number of meals eaten per day	2.77	2.66	0.11
	(0.09)	(0.11)	
Reduced Coping Strategies Index (rCSI)	19.86	18.50	1.36**
	(0.45)	(0.46)	
Household engaged in stress coping strategies	0.85	0.87	-0.02
	(0.01)	(0.01)	
Household engaged in crisis coping strategies	0.59	0.65	-0.06**
	(0.02)	(0.02)	
Household engaged in emergency coping strategies	0.21	0.70	-0.50***
	(0.01)	(0.02)	
Total monthly food expenditure (average)	254.98	295.29	-40.31***
	(8.10)	(10.28)	
Total monthly non-food expenditure (average)	102.80	72.05	30.75***
	(4.48)	(3.51)	
Monthly total household expenditure	357.78	367.34	-9.56
	(10.34)	(11.75)	
Monthly total household expenditure per capita	101.39	97.94	3.45
	(3.42)	(3.72)	
Food Expenditure Share	0.71	0.79	-0.08***
	(0.01)	(0.01)	
Household being vulnerable using FES CARI	0.76	0.92	-0.17***
	(0.02)	(0.01)	
Household being vulnerable using Income rCARI	0.64	0.73	-0.09***
	(0.02)	(0.02)	
Number of observations	760	656	1416

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table 6: Impact of Survey Length in F2F Survey: Long vs. Short

Variable	(1) Total Mean/(SE)	(2) Long Mean/(SE)	(3) Short Mean/(SE)	(2)-(3) Pairwise t-test Mean difference
Consumption over the past 7 days (cereals and tubers)	5.38	5.38	5.38	0.00
	(0.08)	(0.12)	(0.11)	
Consumption over the past 7 days (pulses)	2.00	1.91	2.09	-0.18
	(0.07)	(0.11)	(0.10)	
Consumption over the past 7 days (dairy products)	1.38	1.35	1.40	-0.06
	(0.07)	(0.10)	(0.11)	
Consumption over the past 7 days (protein-rich foods)	3.03	3.06	3.00	0.06
	(0.09)	(0.13)	(0.12)	
Consumption over the past 7 days (vegetables)	3.93	4.00	3.87	0.13
	(0.10)	(0.14)	(0.15)	
Consumption over the past 7 days (fruit)	2.58	2.77	2.40	0.38**
	(0.09)	(0.13)	(0.13)	
Consumption over the past 7 days (oil)	5.38	5.59	5.17	0.42**
	(0.09)	(0.12)	(0.13)	
Consumption over the past 7 days (sugar)	5.39	5.45	5.32	0.14
	(0.09)	(0.13)	(0.13)	
Food Consumption Score	46.28	46.42	46.16	0.26
	(0.78)	(1.11)	(1.10)	
Relied on less preferred, less expensive food	4.41	4.26	4.55	-0.29
	(0.09)	(0.13)	(0.13)	
Borrowed food or relied on help from friends or relatives	1.64	1.63	1.64	-0.01
	(0.07)	(0.10)	(0.10)	
Reduced portion size of meals at meals time	3.80	3.78	3.81	-0.03
	(0.10)	(0.14)	(0.14)	
Restricted Coping by adults in order for young-children to eat	1.87	1.87	1.87	0.00
	(0.09)	(0.13)	(0.12)	
Reduced the number of meals eaten per day	2.77	2.78	2.76	0.02
	(0.09)	(0.13)	(0.13)	
Reduced Coping Strategies Index (rCSI)	19.86	19.69	20.02	-0.32
	(0.45)	(0.66)	(0.62)	
Household engaged in stress coping strategies	0.85	0.82	0.88	-0.06**
	(0.01)	(0.02)	(0.02)	
Household engaged in crisis coping strategies	0.59	0.59	0.60	-0.01
	(0.02)	(0.03)	(0.03)	
Household engaged in emergency coping strategies	0.21	0.18	0.23	-0.05
	(0.01)	(0.02)	(0.02)	
Total monthly food expenditure (average)	254.98	276.03	234.25	41.77***
	(8.10)	(12.26)	(10.51)	
Total monthly non-food expenditure (average)	102.80	112.94	92.82	20.13**
	(4.48)	(7.71)	(4.58)	
Monthly total household expenditure	357.78	388.97	327.07	61.90***
	(10.34)	(16.25)	(12.67)	
Monthly total household expenditure per capita	101.39	112.17	90.78	21.40***
	(3.42)	(5.50)	(4.02)	
Food Expenditure Share	0.71	0.72	0.70	0.02
	(0.01)	(0.01)	(0.01)	
Household being vulnerable using FES CARI	0.76	0.76	0.75	0.01
	(0.02)	(0.02)	(0.02)	
Household being vulnerable using Income rCARI	0.64	0.64	0.64	0.00
	(0.02)	(0.02)	(0.02)	
Number of observations	760	377	383	760

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

Table 7: Impact of Survey Length in Remote Survey: Long vs. Short

Variable	(1) Total Mean/(SE)	(2) Long Mean/(SE)	(3) Short Mean/(SE)	(2)-(3) Pairwise t-test Mean difference
Consumption over the past 7 days (cereals and tubers)	4.93	4.86	4.99	-0.13
	(0.09)	(0.13)	(0.12)	
Consumption over the past 7 days (pulses)	2.50	2.62	2.38	0.24
	(0.09)	(0.13)	(0.12)	
Consumption over the past 7 days (dairy products)	2.03	2.16	1.90	0.26
	(0.09)	(0.13)	(0.12)	
Consumption over the past 7 days (protein-rich foods)	3.48	3.54	3.41	0.13
	(0.09)	(0.13)	(0.13)	
Consumption over the past 7 days (vegetables)	4.75	4.92	4.58	0.34*
	(0.10)	(0.14)	(0.14)	
Consumption over the past 7 days (fruit)	2.90	2.88	2.92	-0.04
	(0.10)	(0.14)	(0.15)	
Consumption over the past 7 days (oil)	5.40	5.51	5.29	0.22
	(0.09)	(0.13)	(0.13)	
Consumption over the past 7 days (sugar)	5.64	5.72	5.56	0.15
	(0.09)	(0.12)	(0.13)	
Food Consumption Score	52.54	53.80	51.30	2.50
	(0.94)	(1.34)	(1.32)	
Relied on less preferred, less expensive food	4.56	4.52	4.60	-0.08
	(0.11)	(0.15)	(0.15)	
Borrowed food or relied on help from friends or relatives	1.75	1.74	1.75	-0.01
	(0.08)	(0.12)	(0.12)	
Reduced portion size of meals at meals time	3.27	3.45	3.09	0.36
	(0.11)	(0.16)	(0.15)	
Restricted Coping by adults in order for young-children to eat	1.51	1.53	1.49	0.04
	(0.09)	(0.13)	(0.12)	
Reduced the number of meals eaten per day	2.66	2.76	2.56	0.20
	(0.11)	(0.15)	(0.14)	
Reduced Coping Strategies Index (rCSI)	18.50	18.79	18.22	0.58
	(0.46)	(0.67)	(0.62)	
Household engaged in stress coping strategies	0.87	0.87	0.88	-0.01
	(0.01)	(0.02)	(0.02)	
Household engaged in crisis coping strategies	0.65	0.66	0.64	0.02
	(0.02)	(0.03)	(0.03)	
Household engaged in emergency coping strategies	0.70	0.70	0.70	0.00
	(0.02)	(0.03)	(0.03)	
Total monthly food expenditure (average)	295.29	391.11	201.78	189.33***
	(10.28)	(17.60)	(8.04)	
Total monthly non-food expenditure (average)	72.05	88.59	55.91	32.67***
	(3.51)	(4.45)	(5.26)	
Monthly total household expenditure	367.34	479.69	257.69	222.00***
	(11.75)	(19.39)	(10.40)	
Monthly total household expenditure per capita	97.94	124.91	71.62	53.29***
	(3.72)	(6.23)	(3.58)	
Food Expenditure Share	0.79	0.78	0.80	-0.02*
	(0.01)	(0.01)	(0.01)	
Household being vulnerable using FES CARI	0.92	0.93	0.92	0.02
	(0.01)	(0.01)	(0.02)	
Household being vulnerable using Income rCARI	0.73	0.71	0.75	-0.05
	(0.02)	(0.03)	(0.02)	
Number of observations	656	324	332	656

Note: Significance: * p<0.10, ** p<0.05, *** p<0.01.

4 Outlier Checks

Table 8: Outlier Check for FCS

	Face-to-Face				Remote			
	Mean	SD	Range	Coef of Variation	Mean	SD	Range	Coef of Variation
Food Consumption Score:								
FCSStap	5.38	2.28	7	0.42	4.93	2.32	7	0.47
FCSPulse	2.00	2.06	7	1.03	2.50	2.24	7	0.90
FCSDairy	1.38	2.02	7	1.47	2.03	2.32	7	1.14
FCSPr	3.03	2.42	7	0.80	3.48	2.39	7	0.69
FCSVeg	3.93	2.82	7	0.72	4.75	2.56	7	0.54
FCSFruit	2.58	2.57	7	0.99	2.90	2.61	7	0.90
FCSFat	5.38	2.52	7	0.47	5.40	2.34	7	0.43
FCSSugar	5.39	2.57	7	0.48	5.64	2.25	7	0.40
FCS	46.28	21.57	112	0.47	52.54	24.12	108	0.46
Reduced Coping Strategy Index:								
rCSILessQlty	4.41	2.58	7	0.59	4.56	2.71	7	0.59
rCSIBorrow	1.64	1.95	7	1.19	1.75	2.13	7	1.22
rCSIMealSize	3.80	2.70	7	0.71	3.27	2.85	7	0.87
rCSIMealAdult	1.87	2.38	7	1.28	1.51	2.24	7	1.48
rCSIMealNb	2.77	2.52	7	0.91	2.66	2.70	7	1.01
rCSI	19.86	12.41	56	0.62	18.50	11.68	56	0.63
Livelihood Coping Strategies:								
Lcs_Stress_Coping	0.85	0.36	1	0.42	0.87	0.33	1	0.38
Lcs_Crisis_Coping	0.59	0.49	1	0.83	0.65	0.48	1	0.73
Lcs_Emergency_Coping	0.21	0.41	1	1.96	0.70	0.46	1	0.65
Expenditure:								
HHExpFoodTotal_1M	254.98	223.18	1670	0.88	295.29	263.29	2336	0.89
HHExpNFTotal_1M	102.80	123.50	1250	1.20	72.05	89.90	1300	1.25
HHExpTotal	357.78	284.94	1828	0.80	367.34	300.91	2352	0.82
PCExpTotal	101.39	94.20	741	0.93	97.94	95.22	999	0.97
FES	0.71	0.19	1	0.26	0.79	0.16	1	0.20
HH_Vul_FES	0.76	0.43	1	0.57	0.92	0.27	1	0.29
Income:								
HH_Vul_Income	0.64	0.48	1	0.75	0.73	0.44	1	0.61